

Qualitative ,Quantitative assay of Zn solubilization:

1. 3.2.4 Screening the isolates for Zn solubilization

Bacterial isolates, obtained from soil sample, were **spot inoculated** on a basal medium supplemented with **0.2% insoluble Zn salts** either ZnO or ZnCO₃ (Appendix 8, Saravanan et al., 2004b). A **loop full** (10 µL) of bacterial culture grown to log phase was **spot inoculated** on the plates and the plates were **incubated at 30 °C for 48 hours**. The **zone of solubilization/clearance** was **observed for 5 days** and the **diameter** of zone of solubilization was **measured and recorded** in mm. Strains solubilizing Zn (either ZnO or ZnCO₃) were selected for further study. Zn solubilization efficiency (ZSE) and Zn solubilization index (ZSI) were calculated using standard formula as given below (Nguyen et al., 1992; Premono et al., 1996).

ZSE= Diameter of solubilization halo zone/ diameter of colony × 100

ZSI = (Diameter of solubilization halo zone + diameter of colony)/colony diameter

Strains showing Zn solubilization efficiency of more than 140% were identified as effective Zn solubilizers.

3.2.5 Quantitative assessment of Zn solubilization by strains using Atomic Absorption Spectroscopy

For quantitative assessment of Zn solubilization, basal medium broth was prepared to which **0.1% insoluble Zn oxide (ZnO) was added**. Each flask containing **50 mL of broth** was inoculated with **1 mL of overnight-grown culture** and the broths were incubated at **30 °C for 10 days on a shaker** set at **150 rpm**. **Samples were withdrawn** after **2, 5, and 10** days of incubation, and **10 mL of the collected samples** were **centrifuged at 10,000 rpm** for **10 minutes**.

Subsequently, **1 mL of the supernatant was transferred to a 50 mL flask, and distilled water was added to bring the volume to 50 mL**. This sample was then used to quantify the available Zn using atomic absorption spectrophotometer (AAS) (GBC 932 Plus, GBC Scientific Equipment Ltd., Melbourne, Australia).

Medium without culture was used as **control**. The **pH of each culture and control** was determined **using a pH meter** (LMPH-9, Labman, India). The **experiment was conducted in triplicates**.

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2. 2.4. Quantitative assay for zinc solubilization in broth media 🍏

The quantitative Zn solubilizing potential of the bacterial isolates was determined in a liquid basal medium following the method of Saravanan et al. (2007). The bacterial isolates (1 mL overnight grown culture at 30 ° C, 120 rpm) were inoculated into 100 mL basal medium broth containing 0.1 % of insoluble Zn compounds [ZnO, ZnCO₃, Zn₃(PO₄)₂ and] added separately. Each treatment was subjected to three replications and incubated at 30 ° incubator at 120 rpm for **ten days**. A flask containing an uninoculated medium with insoluble Zn sources was kept as a control during the experiment. The samples were processed at intervals on the **3rd, 5th, 7th, and 10th days**. Subsequently, these were aseptically transferred to centrifuge tubes and subjected to centrifugation at 8000 rpm for 10 min. Following the infiltration of the supernatants through a 0.45µ mfilter, a 1-mL filtrate was diluted with sterile distilled water to a total volume of 50 mL and subsequently utilized for estimating the soluble Zn content using Atomic Absorption Spectrophotometry (AAS).

2.5. pH measurement during the process of zinc solubilization

In addition to the quantitative assessment of Zn solubilization, **pH variations in the culture broth and uninoculated medium were estimated on the 3rd, 5th, 7th, and 10th days of incubation**. To achieve this, the **pH of the basal medium was adjusted to 7.0 by adding 0.1 M HCl and NaOH as needed**. Aliquots of the samples were subjected to **centrifugation**, and subsequently, the **pH of the resulting supernatant was measured using a pH meter** (Bhakat et al., 2021). **Pearson product-moment correlation analysis program** was used to observe the correlation within the Zn solubilization index, the quantity of Zn solubilized by the bacterial isolates, and the corresponding pH reduction of the culture medium.

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[3.3.1 Physicochemical analysis of soil samples

3.2.2 Physicochemical analysis of soil samples

Physicochemical analysis of soil samples including soil pH, electrical conductivity, organic carbon, Available Nitrogen, Phosphorous, Potassium, Sulphur and micronutrient concentration including Zn, Mn, Cu and Fe were detected using AAS, additionally moisture percentage and moisture correction factor were determined using standard methods described by Gupta (Gupta, 2007).

result:

Details of soil samples collected for the study are given in table 3.2. The soil samples

were analyzed for their physicochemical properties. The pH of the soil sample was in a range of 7.22 to 7.80. The electrical conductivity (EC) (mS/cm) of soil samples varied from 0.13 to 0.29 mS/cm in the collected soil sample. A maximum organic carbon content of 0.81% was found in sample PP collected from a field in the faculty of Agricultural Sciences, Aligarh Muslim University, Aligarh. The highest available nitrogen (N), and phosphorous (P) was found in sample S1 and W3 respectively, with values in the range of 269.59 - 27.72 Kg/ha respectively. Whereas a high concentration of potassium (K) i.e. 292.50 Kg/ha was found in sample VF and S1 (Table 3.2). The highest concentration of sulphur (S) 1.61 ppm was detected in the sample PP.

The concentration of various metals and elements in soil was detected using atomic absorption spectrometry (AAS). The analysis revealed the collected soil samples were Zn deficient with a DTPA extracted Zn concentration in the range of 0.51- 0.55 ppm, except for samples PP, S1 and W2 where the concentration was 2.22, 1.99 and 0.76 ppm, respectively. All the soils were deficient for manganese (Mn) having concentration of less than 2.0 ppm. The highest concentration of sulphur (S) and copper (Cu) i.e. 11.82 and 1.61 ppm respectively was found in sample PP. Whereas, highest concentration of iron (Fe, 6.29 ppm) was found in sample S1 collected from the agricultural land of badaun (Table 3.2)]